

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology & Geophysics

2. Subject Name: Paleontology & Stratigraphy **L-T-P:** 3-0-0 **Credits:** 3

3. Pre-requisites: NA

4. Syllabus and reference books:

Syllabus:

Paleontology: Definition of fossils, taphonomic processes. Taxonomy. Distinctive features of morphology, ecology, evolution and geological significance of major animal phyla.

Stratigraphy: Stratigraphic principles. Stratigraphic schemes. The Archean, Proterozoic and Phanerozoic geology of India.

Reference Books:

- 1) M. Ramakrishnan and R. Vaidyanathan, "Geology of India," Geological Society of India, Bangalore, Vol. 1, 2008.
- 2) M. Ramakrishnan and R. Vaidyanathan, "Geology of India," Geological Society of India, Bangalore, Vol. 2, 2008
- 3) Principles of Sedimentology and Stratigraphy, Sam Boggs Jr., Pearson; 5th edition
- 4) Sedimentology and Stratigraphy, Gary Nichols, Wiley-Blackwell.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1	Fossils & fossilization processes	4
2	Taxonomy	2
3	Morphology, ecology, evolution and geological significance of major animal phyla	10
4	Stratigraphic principles	4
5	Stratigraphic schemes	10
6	Geology of India	8
Total number of hours		38